

[Geographic atrophy imaging using fundus autofluorescence method].

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Geographic atrophy is a manifestation of the advanced age-related macular degeneration and form of irreversible atrophy of retinal pigment epithelium and photoreceptor layer. Early detection of changes and the ability to evaluate disease progression accurately constitute a key problem in diagnosis and treatment planning. Fundus autofluorescence is a relatively new imaging method considered nowadays to be the best in diagnosis and observing the natural or treatment-altered course of disease. High resolution images showing the 3D distribution of retinal pigment epithelium autofluorescence as lipofuscin index can be obtained owing to the launch of the confocal scanning laser ophthalmoscope.

Geographic atrophy - Signs, symptoms, and quality of life.

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Geographic atrophy (GA) is a prevalent cause of vision loss among elderly and is associated with a significant loss of function. We reviewed the current literature to assess the effect of GA on patients' daily lives and well-being. We record and organize the signs, symptoms, and impacts that are important in life with GA. Further, we examined the impact of GA on vision-related quality of life. The main complaint among patients was difficulties regarding daily tasks, especially reading and other near activities. However, a large proportion of patients also reported fear, frustration, and anxiety as salient symptoms with large impact. Many patients do not have adequate information about their condition as well as the prognosis. The most commonly used measure of patient-reported outcome measure (PROM) is the National Eye Institute Visual Function Questionnaire (VFQ), that reflects the severity of impact on 12 subscales, from where near activities, general vision, mental health, and role difficulties had the lowest scores. Longitudinal studies of GA and the impact of low-vision rehabilitation efforts on health-related quality of life are sparse but suggest a significant improvement on several items. PROM is included in clinical trials, and so far, no drug has shown to improve the functional outcome in terms of PROM.

Future perspectives for treating patients with geographic atrophy.

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Geographic atrophy (GA) is a late-stage form of age-related macular degeneration (AMD) characterized by the expansion of atrophic lesions in the outer retina. There are currently no approved pharmacological treatments to prevent or slow the progression of GA. This review describes the progression and assessment of GA, predictive imaging features, and complement-targeting investigational drugs for GA. A literature search on GA was conducted. Expansion of atrophic lesions in patients with GA is associated with a decline in several measures of visual function. GA lesion size has been moderately associated with measures obtained through microperimetry, whereas GA lesion size in the 1-mm diameter area centered on the fovea has been associated with visual acuity. Optical coherence tomography (OCT) can provide 3-dimensional quantitative assessment of atrophy and is useful for identifying early atrophy in GA. Features that have been found to predict the development of GA include certain drusen characteristics and pigmentary abnormalities. Specific OCT features, including hyper-reflective foci and OCT-reflective drusen substructures, have been associated with AMD disease progression. Lesion characteristics,

including focality, regularity of shape, location, and perilesional fundus autofluorescence patterns, have been identified as predictors of faster GA lesion growth. Certain investigational complement-targeting drugs have shown efficacy in slowing the progression of GA. GA is a progressive disease associated with irreversible vision loss. Therefore, the lack of treatment options presents a significant unmet need. OCT and drugs under investigation for GA are promising future tools for disease management.

Subretinal Transplantation of Human Embryonic Stem Cell Derived-retinal Pigment Epithelial Cells into a Large-eyed Model of Geographic Atrophy.

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Geographic atrophy (GA), the late stage of dry age-related macular degeneration is characterized by loss of the retinal pigment epithelial (RPE) layer, which leads to subsequent degeneration of vital retinal structures (e.g., photoreceptors) causing severe vision impairment. Similarly, RPE-loss and decrease in visual acuity is seen in long-term follow up of patients with advanced wet age-related macular degeneration (AMD) receiving intravitreal anti-vascular endothelial growth factor (VEGF) treatment. Therefore, on the one hand, it is fundamental to efficiently derive RPE cells from an unlimited source that could serve as replacement therapy. On the other hand, it is important to assess the behavior and integration of the derived cells in a model of the disease entailing surgical and imaging methods as close as possible to those applied in humans. Here, we provide a detailed protocol based on our previous publications that describes the generation of a preclinical model of GA using the albino rabbit eye, for evaluation of the human embryonic stem cell derived retinal pigment epithelial cells (hESC-RPE) in a clinically relevant setting. Differentiated hESC-RPE are transplanted into naive eyes or eyes with NaIO₃-induced GA-like retinal degeneration using a 25 G transvitreal pars plana technique. Evaluation of degenerated and transplanted areas is performed by multimodal high-resolution non-invasive real-time imaging.

En Face Optical Coherence Tomography Imaging for the Detection of Nascent Geographic Atrophy.

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To determine if en face optical coherence tomography (OCT) imaging can identify nascent geographic atrophy (nGA) in eyes with intermediate age-related macular degeneration (iAMD). Retrospective observational case series. Patients with iAMD from the COMPLETE study at the Bascom Palmer Eye Institute were evaluated to determine if nGA was present at baseline and at follow-up using high-density Spectralis OCT B-scans and en face OCT images from the Cirrus OCT instrument. If available, additional en face OCT images and B-scans were analyzed at follow-up times beyond the 52-week period. A total of 37 eyes (27 patients) were evaluated for at least 1 year using both B-scans and en face images. Two drusen suspicious for nGA at baseline were identified, but neither druse developed GA after 24 and 62 months of follow-up, respectively. Another druse displayed hypertransmission into the choroid at week 52 on B-scan imaging and was classified as nGA. En face OCT imaging identified this druse as a focal bright area. Drusen breakdown occurred during a follow-up of 39 months. En face OCT imaging appeared to be as useful as routine B-scan imaging for identifying areas suspicious for nGA in this population from the COMPLETE Study. Additional longitudinal follow-up of eyes with drusen is needed to determine if en

face OCT imaging can replace the evaluation of individual B-scans for the detection of nGA.

Characteristics of Hyperautofluorescent Choroidal Vessels within the Macular Atrophic Area Using Spectral-Domain Optical Coherence Tomography.

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The aim of this study was to clarify the characteristic findings in patients with geographic atrophy with or without hyperautofluorescent choroidal vessels within macular atrophic areas on short-wavelength fundus autofluorescence imaging. Sixty-seven eyes of 43 consecutive patients with macular atrophic areas were divided into groups with (group 1) and without (group 2) hyperautofluorescent choroidal vessels on fundus autofluorescence imaging and then retrospectively studied using spectral-domain optical coherence tomography. In group 1 ($n = 21$), the average subfoveal choroidal thickness was $61.5 \pm 20.1 \mu\text{m}$, and the average foveal retinal thickness was $93.0 \pm 51.3 \mu\text{m}$. On the other hand, in group 2 ($n = 46$), the average subfoveal choroidal thickness was $200.7 \pm 83.1 \mu\text{m}$, and the average foveal retinal thickness was $109.2 \pm 58.5 \mu\text{m}$. Although retinal thickness did not differ significantly between the two groups ($p = 0.28$), the difference in choroidal thickness was statistically significant ($p < 0.001$). Choroidal thinning might contribute to the hyperautofluorescence of choroidal vessels.

Characteristics of Geographic atrophy in an elderly UK population-The Bridlington Eye Assessment Project (BEAP): a cross-sectional study (2002-2006).

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Geographic atrophy (GA) is a common cause of visual loss. The UK population prevalence is unknown. We studied GA prevalence, characteristics, and associations in an elderly UK population. Masked grading of colour fundus photographs from 3549 participants in the cross-sectional study of Bridlington residents aged ≥ 65 years. GA size, shape and foveal involvement were correlated with demography and vision. GA was detected in 130 eyes (101 individuals) of 3480 participants with gradable images (prevalence 2.90%; 95% CI 2.39-3.52 either eye), was bilateral in 29/3252 subjects (0.89%, 95% CI 0.62-1.28) with bilateral gradable photos, with mean age of 79.26 years (SD 6.99, range 67-96). Prevalence increased with age, from 1.29% (95% CI 0.69-2.33) at 65-69 to 11.96% (95% CI 7.97-17.50) at 85-90 years. Mean GA area was 4.51 mm^2 (SD 6.48, 95% CI 3.35-5.66); lesions were multifocal in 47/130 eyes (36.2%; 95% CI 28.4-44.7). Foveal involvement occurred in 41/130 eyes (31.5%; 95% CI 24.2-40.0). In eccentric GA, mean distance from circumference to fovea was $671 \mu\text{m}$ (SD 463; 95% CI 570-773). Older age (OR 1.10/year increase; 95% CI 1.06-1.14), RPD (OR 1.87; 95% CI 1.10-3.19) and large drusen/RPD $\geq 125 \mu\text{m}$ (OR 6.16; 95% CI 3.51-10.75) were significantly associated with GA in multivariate analysis. GA lesions (18/31 eyes; 58%; 95% CI 40.7-73.6) had U-shape configuration more frequently in RPD subjects than those without (9/99 eyes, 9.1%; 95% CI 4.66-16.6) ($p = 0.0001$). GA, commonly solitary and eccentric, occurred in the perifovea. However, one third of GA eyes had foveal and bilateral involvement. Possible association of RPD with GA phenotype exists. Population multimodal imaging studies may improve understanding further.

A Review of Current and Future Management of Geographic Atrophy.

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Age-related macular degeneration (AMD) is a multifactorial disease and a leading cause of vision impairment in elderly people in Western society. Geographic atrophy (GA), the late stage of dry AMD, is typically defined as a round or oval area of atrophy of 175 μm or more. In GA patients, visual acuity (VA) can still be good if the macula is spared, but decreased if GA extends through the fovea causing a great impairment of quality of life. Because of a poor correlation between VA and GA lesions or progression, a multimodal imaging approach is necessary to better follow up GA patients. In the last years, the introduction in clinical practice of new non-invasive tools such as fundus autofluorescence, structural optical coherence tomography (OCT) and OCT angiography helped the ophthalmologists to better understand the natural course of GA patients. However, several pathways concerning the pathogenesis of the disease are not completely clarified yet and should be investigated further. Although no approved therapy exists for GA, healthy lifestyle and nutritional intervention with some specific supplementations (e.g., vitamins C and E, beta-carotene, high dietary folate) may help to prevent the onset and to delay the progression of the disease. At the same time, several drugs are under evaluation in clinical trials with interesting results. These drugs try to stop several pathways implicated in the pathogenesis of GA, but probably only a few of these will prove truly effective, confirming the preliminary results, and will be available in clinical practice.

An Acute Injury Model for the Phenotypic Characteristics of Geographic Atrophy.

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Geographic atrophy (GA) is the late stage of non-neovascular age-related macular degeneration. A lack of animal models for GA has hampered treatment efforts. Presented herein is a rat model for GA using subretinal injection of sodium iodate (NaIO_3). Rats were given subretinal injections of NaIO_3 (5 $\mu\text{g}/\mu\text{L}$) using a pico-injector. Fundus photographs and spectral domain optical coherent tomography scans were collected at 1, 3, 7, 14, and 28 days after injection, at which time rats were euthanized and eyes were enucleated. Eyes were either cryopreserved or dissected into retinal and choroidal flatmounts. Fluorescence immunohistochemistry was performed for retinal glial fibrillary acidic protein (activated Müller cells and astrocytes) and vimentin (Müller cells), as well as peanut agglutinin lectin (photoreceptors) labeling. RPE/choroids were labeled for RPE65 and CD34. Images were collected on Zeiss confocal microscopes. Fundus photos, spectral domain optical coherent tomography, and RPE65 staining revealed well-demarcated areas with focal loss of RPE and photoreceptors in NaIO_3 -treated rats. At 1 day after injection, RPE cells appeared normal. By 3 days, there was patchy RPE and photoreceptor loss in the injected area. RPE and photoreceptors were completely degenerated in the injected area by 7 days. A large subretinal glial membrane occupied the degenerated area. Choriocapillaris was highly attenuated in the injected area at 14 and 28 days. The rat model reported herein mimics the photoreceptor cell loss, RPE atrophy, glial membrane formation, and choriocapillaris degeneration seen in GA. This model will be valuable for developing and testing drugs and progenitor cell regenerative therapies for GA.

Investigational Agents in Development for the Treatment of Geographic Atrophy Secondary to Age-Related Macular Degeneration.

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Geographic atrophy (GA) is an advanced form of age-related macular degeneration, a late-onset, complex, genetic degenerative disease of the retina. Multiple environmental and genetic factors have been implicated in the development of GA, a pathology ultimately defined by loss of photoreceptors and the underlying retinal pigment epithelium and choriocapillaris. The personal burden of GA has been documented to have a substantial negative impact on quality of life, with progressive and cumulative loss of visual function each year. Currently, there are no treatments to prevent or slow the development or progression of GA. Multiple genetic and histopathologic studies have implicated dysregulation of the complement cascade in GA pathogenesis, leading to the development of several investigational pharmaceuticals targeting key factors in this inflammatory pathway. Several other biochemical pathways have also been implicated in GA development and progression, such as mitochondrial components, mediators of apoptosis and molecules involved in extracellular matrix remodeling, many of which are also being investigated for their potential value as therapeutic targets for GA treatment. Recent advancements in our understanding of GA pathogenesis and the progression of multiple potential therapeutics into later-stage human clinical trials hold great promise for a clinically effective therapeutic for patients with GA to emerge within the near future.
